

9/7/2023

Maths

① Monomial

$$\textcircled{2} (x+1)(x^2 - x - x^4 + 1)$$

$$x^3 - x^2 - x^5 + x + x^2 - x - x^4 + 1$$

Five is the highest power and the degree of the polynomial

$$-x^5 = \text{degree } 5$$

$$\textcircled{3} a^2x^2 - 3ax + 3x - 1 = 0$$

$$f(1) = a^2(1)^2 - 3a(1) + 3(1) - 1 = 0$$

$$= a^2 - 3a + 3 - 1 = 0$$

$$a^2 - 3a + 2 = 0$$

$$a^2 - 1a - 2a + 2$$

$$a(a-1) - 2(a-1)$$

$$(a-2)(a-1)$$

$$a=2, a=1$$

$\therefore$ Value of a is 2, 1

④ Zeros of the graph - no. of times it crosses x axis - 1

⑤ if the zeros are α, β then

$$\alpha, \beta = \frac{1}{\alpha}$$

$\therefore$ the zeros are $\alpha, \frac{1}{\alpha}$

$$x^2 - (\text{sum of zeros}) + (\text{product of zeros})$$

$$\alpha + \frac{1}{\alpha} = -(-3)$$

$$\alpha \times \frac{1}{\alpha} = K$$

$$\frac{x}{x} = \frac{k}{x}$$

$$\boxed{k=1}$$

6) we know that $= x^2 - (\text{sum of roots}) + (\text{product of roots})$

$$\alpha + \beta = 5$$

$$\alpha\beta = -14$$

$$= x^2 - |5| + (-14)$$

$$x^2 - 5 - 14$$

$$7) x^2 + 5x + 6$$

$$= x^2 + 3x + 2x + 6$$

$$x(x+3) + 2(x+3)$$

$$(x+2)(x+3)$$

$$x = -2, x = -3$$

$$\alpha + \beta = -\frac{b}{a} = -\frac{-4}{3} = \frac{4}{3} \quad \text{LHS} = \text{RHS}$$

$$\alpha\beta = \frac{c}{a} = \frac{-6}{3} = -2 \quad \text{LHS} = \text{RHS}$$

8)

$$9) x + \frac{1}{x} = 3$$

$$x^3 + \frac{1}{x^3} = ?$$

Let $x = a, \frac{1}{x} = b$

$$a^3 + b^3 = (a+b)^3 - 3ab(a+b)$$

$$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x\right)\left(\frac{1}{x}\right)\left(x + \frac{1}{x}\right)$$

$$x^3 + \frac{1}{x^3} = (3)^3 - 3 \times \frac{1}{x} (3) \left[\text{we know that } x + \frac{1}{x} = 3 \right]$$

$$= 27 - 9$$

$$\therefore x^3 + \frac{1}{x^3} = 18$$

(10) if α and β are zeros, then -
 $[x^2 - (\text{sum of zeros}) + (\text{product of zeros})]$

$$\alpha + \beta = -(-5)$$

$$\alpha\beta = 6$$

$$\left(1 + \frac{1}{\alpha}\right) \left(1 + \frac{1}{\beta}\right)$$

$$= \left(\frac{\alpha+1}{\alpha}\right) \left(\frac{\beta+1}{\beta}\right)$$

$$= \frac{\alpha\beta + \alpha + \beta + 1}{\alpha\beta}$$

= substituting the values

$$= \frac{6 + 5 + 1}{6}$$

$$= \frac{12}{6} = 2$$

$$\left(1 + \frac{1}{\alpha}\right) \left(1 + \frac{1}{\beta}\right) = 2$$